International Institute of Information Technology, Hyderabad

(Deemed to be University)

SC1.308: The Universe Across Scales
Question cum Answer Booklet
End-Semester Examination

50

Max. Time: 2 Hrs **Total Marks: 50**

Roll No: 2022111004 **Programme:** CSD **Date of Exam:** 29 Apr 2025

Room no: H204 **Seat No:** B8. **Invigilator's Signature:**

Special Instructions about the exam

1. You may bring **one handwritten A4 size cheat sheet** but **no printed/photocopied material is allowed.** You must submit the cheat sheet with this booklet.
2. Calculators are allowed, but no other electronic gadgets (laptops, tablets, mobiles, etc.) are permitted.
3. There are total **10** questions, attempt all of them.
4. Additional sheets for rough work are allowed, but those must be submitted with this booklet.

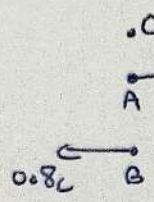
Marks Table (To be filled by the Examiner)

Question No	CO numbers							Examiner
1	2, 5	5						
2	1, 5	5						
3	1, 5	5						
4	1, 2, 5	5						
5	1, 5	5						
6	2, 5	5						
7	1, 5	5						
8	2, 5	5						
9	1, 5	5						
10	2, 5	5						

General Instructions to the students

1. Place your Permanent / Temporary Student ID card on the desk for verification by the invigilator during the examination.
2. Reading materials such as books are not allowed inside the examination hall.
3. Borrowing writing materials or calculators from other students in the examination hall is prohibited.
4. If any student is found indulging in malpractice or copying in the examination hall, the student will be given 'F' grade for the course and may be debarred from writing other examinations.

Q1. Suppose you observe two muons going away from each other with velocity $\pm 0.8c\,\hat{z}$ in a lab. What is the relative velocity of the muon moving towards $-\hat{z}$ with respect to the other? [5]



relative vel of B wrt A.

consider O' as the stationary frame.

consider frame of O' as prime frame.

A as ~~O'~~ main frame.

O' moves with $-0.8c$ (v) along z axis. (-ve dir)

B moves with $-0.8c$ wrt A $\Rightarrow u_x'$ $\quad v \to -0.8c$

now $u_x \to$ speed of B rel to A,

$$u_x = \frac{u_x' + v}{1 + \frac{v u_x'}{c^2}} = \frac{-1.6c}{1 + 0.64} = \frac{-1.6c}{1.64} = -0.976c$$

$\therefore$ A sees B moving along $-z$ axis with speed of $0.976c$

$= 2.93 \times 10^8$ m/s

— Ans.

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Q2. List the fermions in the Standard Model and specify the interactions (electromagnetic/weak/strong) that each of these interacts through. [5]

There are 12 fermions of 2 types $\Rightarrow$ 6 quarks + 6 leptons

quarks			leptons		
up	charm	top	e^-	muon	tau
down	strange	bottom	e-neutrino	μ-neutrino	τ-neutrino

quarks interact with each other by gluons - strong interaction

quarks and leptons interact with each other by photons - Electromagnetic

and bozons - weak interaction.

Interactions:

u c t
d s p } gluons - strong.

e μ τ
ν_e ν_μ ν_τ } photon $\to$ EM

bozons $\to$ weak.

interact with everything that contains mass.

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Q3. Write down the kinetic energy operator. Obtain the expected kinetic energy of an electron trapped in a one-dimensional box of length L (assume $V(x) = 0$ inside the box) if it is in the 4th energy state. [2+3]

kinetic energy operator $\hat{T} = \frac{1}{2m}\left(\frac{\hbar}{i}\frac{\partial}{\partial x}\right)^2$ reason: $\hat{p} = \frac{\hbar}{i}\frac{\partial}{\partial x}$ and $T = \frac{p^2}{2m}$

solution in particle in a box: $\psi_n(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{n\pi x}{L}\right)$, $E_n = \frac{n^2\pi^2\hbar^2}{2mL^2}$ — ①

with $\Psi_n(x,t) = \psi_n(x)e^{-iE_nt/\hbar}$

$n=4$.

$\psi_4(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{4\pi x}{L}\right)$

$\langle T\rangle = \int_{-\infty}^{\infty}\psi_4^*\hat{T}\psi_4\,dx$. now $\frac{\partial}{\partial x}\psi_4(x) = \sqrt{\frac{2}{L}}\cos\left(\frac{4\pi x}{L}\right)\cdot\frac{4\pi}{L}$

$\frac{\hbar}{i}\frac{\partial}{\partial x}\psi_4(x) = \frac{4\pi}{L}\sqrt{\frac{2}{L}}\cos\left(\frac{4\pi x}{L}\right)\frac{\hbar}{i}$

$= \int_0^L \psi_4^*\hat{T}\psi_4\,dx$ $\frac{\hbar}{i}\frac{\partial}{\partial x}\left(\frac{\hbar}{i}\frac{\partial}{\partial x}\psi_4(x)\right) = \frac{\hbar^2}{i^2}\frac{4\pi}{L}\sqrt{\frac{2}{L}}\,-\sin\left(\frac{4\pi x}{L}\right)\frac{4\pi}{L}$

since ψ is 0 outside the box.

$= \hbar^2\frac{16\pi^2}{L^2}\sqrt{\frac{2}{L}}\sin\left(\frac{4\pi x}{L}\right)$

$\langle T\rangle = \int_0^L \frac{\hbar^2 16\pi^2}{2mL^2}\frac{2}{L}\sin^2\left(\frac{4\pi x}{L}\right)dx$.

$= \frac{\hbar^2 32\pi^2}{2mL^3}\int_0^L \sin^2\left(\frac{4\pi x}{L}\right)dx = \frac{\hbar^2 32\pi^2}{2mL^3}\times\frac{L}{2}$

$= \frac{16\hbar^2\pi^2}{2mL^2}$ — Ans.

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which matches the original formula ① for $n=4$. — Ans.

Q4. A pion is created near the top of Earth's atmosphere, about 100 km above sea level. It travels vertically downward with a total energy of $E = 1.4 \times 10^5$ MeV. In its rest frame, it decays 35.0 ns after its creation. At what altitude above the sea level, as measured in the Earth's reference frame, does the decay occur? We know that the rest energy of a pion is about 140 MeV. (1 eV = 1.6×10^{-19} J, 1 MeV = 10^6 eV) [5]

In earth reference frame, rest energy = 140 MeV $= m_0c^2$

creation energy (speed v) $m_vc^2 = 1.4\times10^5$ MeV

$\gamma m_0c^2 = 1.4\times10^5$ MeV $\gamma = \frac{1}{\sqrt{1-v^2/c^2}}$

$\Rightarrow \gamma = \frac{1.4\times10^5\text{ MeV}}{140\text{ MeV}} = 10^3$.

$\gamma = 10^3 = \frac{1}{\sqrt{1-v^2/c^2}} \Rightarrow 1-\frac{v^2}{c^2} = 10^{-6}$

$\Rightarrow v^2 = (1-10^{-6})c^2 \Rightarrow v = (1-10^{-6})^{1/2} c$

$= (1-\frac{1}{2}10^{-6})c$ (binomial approx)

$v = \frac{\text{distance}}{\text{time}} \Rightarrow \text{distance} = v \times 35\times10^{-9}\,s \cdot \gamma$

time in earth frame $= \gamma \times 35\,ns$. (time dilation)

$= (1-\frac{10^{-6}}{2}) \times 3\times10^8 \times 35\times10^{-6}$

$= (1-\frac{10^{-6}}{2}) 105\times10^2\ m$

$\sim 10.5\ km$ (less than 10.5 km)

altitude at which decay occurs $= 100 - 10.5$

$= 89.5\ km$

— Ans.

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Q5. What do we mean by stationary-state solutions of the Schrödinger equation? Explain how it is possible to get stationary-state solutions, even though the equation depends on time. [2+3]

when the potential is only dependent on x, it is possible to split the wave function into two separe parts $\psi(x)$ and $\phi(t)$ each governed by their own equation.

$\phi(t)$ solution $\rightarrow$ ~~time independent~~ probability of the stationary states.

stationary states correspond to the eigen functions of the Hamiltonian operator.

say $\Psi(x,t) = \psi(x)\phi(t)$. Schrödingers eq: $i\hbar\frac{\partial\Psi}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\Psi}{\partial x^2} + \underbrace{V}_{V(x)}\Psi$ — Ans.

substitute:

$i\hbar\psi(x)\frac{\partial}{\partial t}\phi(t) = -\frac{\hbar^2}{2m}\phi(t)\frac{\partial^2\psi(x)}{\partial x^2} + V(x)\psi(x)\phi(t)$

divide both sides by $\phi(t)\psi(x)$

$$\frac{i\hbar}{\phi(t)}\frac{d\phi(t)}{dt} = -\frac{\hbar^2}{2m}\frac{\partial^2\psi(x)}{\partial x^2}\frac{1}{\psi(x)} + V(x) = E \text{ constant.}$$

$\therefore$ The time evolution of $\Psi(x,t)$ is independent of the position evolution of $\Psi(x,t)$

$\therefore$ There can be stationary state solutions.

since time and x functions are both equal, they need to be equal to some constant E (ind of time and x)

consider the time-ind SE: $-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}\frac{1}{\psi(x)} + V(x) \rightarrow \hat{H}$ (Hamiltonian)

$\therefore \hat{H}\psi = E\psi \rightarrow$ give the solutions of ψ as energy eigenfunctions

'stationary' $\rightarrow$ independent of time evolution.

— Ans.

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Q6. Expand the relativistic expression of energy when $v \ll c$ keeping terms up to $O(v^2/c^2)$. Interpret the terms. Show, using the velocity transformation relations in special relativity, that the speed of light remains the same in all inertial frames. [2+3]

relativistic expression of energy: $E = \frac{mc^2}{\sqrt{1-v^2/c^2}} = mc^2\left(1-\frac{v^2}{c^2}\right)^{-1/2}$

when $v \ll c$ opening with binomial expansion:

$$E = mc^2\left(1-\left(\frac{-1}{2}\frac{v^2}{c^2}\right)\right) = mc^2\left(1+\frac{v^2}{2c^2}\right)$$

$$= mc^2 + \frac{1}{2}mv^2$$

Energy due to rest mass. kinetic energy of particle.

— Ans.

→ Relativistic velocity addition.

$u_x = \frac{u_x' + v}{1+\frac{vu_x'}{c^2}}$ $u_x \to$ speed of particle seen in O

$u_x' \to$ " " O'

$v \to$ speed of O' rel to O.

O → rest frame

O' → frame moving with v along +x.

say particle is light (photon) $u_x' = c$.

$$u_x = \frac{c+v}{1+\frac{vc}{c^2}} = \frac{(c+v)c^2}{c^2+vc} = \frac{(c+v)c^2}{c(c+v)} = \frac{c^2}{c} = c$$

∴ viewed from frame O gives same speed c as seen in O'.

∴ speed of light is same in all inertial frames.

— Ans.

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Q7. A particle of mass m is trapped inside a one-dimensional region between two infinitely high potential barriers separated by a distance L_0. Determine the minimum energy of the particle using the uncertainty principle. [5]

uncertainity principle $\Delta x \cdot \Delta p \geq \frac{h}{2}$ $\Delta x \to$ uncertainity in position

$\Delta p \to$ " momentum.

also $E = \frac{p^2}{2m}$.

since $\Delta x \leq L_0$ as the maximum length it can be in is L_0 inside the well

$\Delta p \geq \frac{h}{2L_0}$

∴ minimum $p = \frac{h}{2L_0}$

5

$\min E = \frac{p^2}{2m}$ $\quad p \to p_{min}$

$= \frac{h^2}{4L_0^2 \times 2m} = \frac{h^2}{8mL_0^2}$ — Ans.

as $\hbar = \frac{h}{2\pi}$ $\quad \min E = \frac{4\pi^2\hbar^2}{8mL_0^2} = \frac{\pi^2\hbar^2}{2mL_0^2}$ — Ans.

Q8. The relation $E = m_v c^2$ can also be written as $E^2 = A\,c^2 + B\,m^2$. Obtain A and B. **[5]**

m_v = mass of object (with rest mass m) travelling with speed v.

$m_v = \gamma m$ where $\gamma = \frac{1}{\sqrt{1-v^2/c^2}}$ $\Rightarrow \gamma^2 = \frac{c^2}{c^2-v^2}$ — (1)

$\gamma^2 m^2 c^2 (c^2 - v^2)$

$E^2 = (\gamma m c^2)^2 = \frac{1}{(1-\frac{v^2}{c^2})} m^2 c^4 = \frac{m^2c^4}{(c^2-v^2)} = m^2c^2\gamma^2$

$= m^2c^2\gamma^2 \frac{(c^2-v^2)}{c^2\gamma^2} = \frac{m^2c^2\gamma^2c^2}{c^2} + m^2\gamma^4v^2$

from (1)

$\gamma = v^3$

$t = 35 \times 10$

$E^2 = (\gamma m c^2)^2$

~~$E^2 = m^2\gamma^2c^2$ where $c^2 = \gamma^2(c^2-v^2)$~~

$E^2 = m^2c^2\gamma^2 = m^2\gamma^2(c^2-v^2) + m^2\gamma^2v^2$ ~~$(\pm m^2\gamma^2v^2)$~~

from (1) $\gamma^2(c^2-v^2) = c^2$

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$E^2 = m^2c^2 + m^2\gamma^2v^2$

$E^2 = \gamma^2m^2c^4 \quad (\pm \gamma^2m^2v^2c^2)$

$= \gamma^2m^2c^4 - \gamma^2m^2v^2c^2 + \gamma^2m^2v^2c^2$ $\quad (p = \gamma m v)$

$= \gamma^2m^2c^2(c^2-v^2) + p^2c^2$

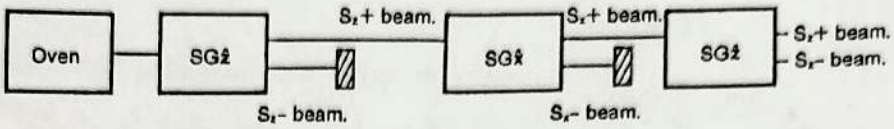
$= m^2c^4 + p^2c^2$ (from (1)

$\therefore E^2 = m^2c^4 + p^2c^2$

$A = p^2 = \frac{m^2v^2}{1-v^2/c^2}$

$B = c^4$ } Ans.

Q9. Two persons, A and B, perform the Stern-Gerlach experiment by passing a beam of electrons through three inhomogeneous magnetic fields and, in between, allowing only the spin-up components to pass through. For A, the magnetic fields point in the $\hat{z}$, $(\hat{z}+\sqrt{3}\hat{y})/2$, and $\hat{z}$ directions sequentially; for B, they point along $\hat{z}$, $\hat{x}$, and $\hat{z}$ directions sequentially. For example, the figure below illustrates B's setup.



Estimate the ratio of the numbers of up-spin measured by A and B if the fluxes of the two beams are identical and they measure for an extended but equal amount of time. [5]

originally 1 e^- started (deal with probability)

P(up spin remaining) $\cos^2\frac{\theta}{2}$ θ is angle rotated.

~~After~~ since original e^- could be randomly oriented,

S_z+ beam (after first detector) has 0.5 e^- for both A and B.

For B → $\cos\theta = \hat{z}\cdot\hat{x} = 0 \Rightarrow \theta = 90^\circ \Rightarrow \cos^2\frac{\theta}{2} = \frac{1}{2}$

After second detector B has $\frac{1}{4}e^-$.

For third magnetic field, $\theta = 90^\circ \rightarrow \cos^2\theta/2 = \frac{1}{2} \Rightarrow$ remaining with $\frac{1}{8}e^-$ — Ans.

For A → $\cos\theta = \frac{\hat{z}}{2}(\hat{z}+\sqrt{3}\hat{y}) = \frac{1}{2}+0 \Rightarrow \theta = 60^\circ$

$\cos^2\theta/2 = 3/4$.

After second magnetic field A has $\frac{1}{2}\times\frac{3}{4} = \frac{3}{8}e^-$

For third MF, $\cos\theta = \frac{1}{2}$ $\therefore \cos^2\theta/2 = 3/4$.

End → A has $\frac{3}{8}\times\frac{3}{4} = \frac{9}{32}e^-$ — Ans.

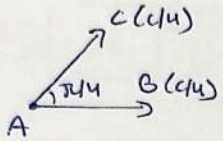
Final ratio of upspins $= \frac{n_A}{n_B} = \frac{9}{32}\times\frac{8}{1} = \frac{9}{4}$ — Ans.

9:4

$\frac{1\times\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}}{1\times\frac{1}{2}\times\frac{3}{4}\times\frac{3}{4}} = 4:9$

Q10. Imagine three ultra-fast jet planes, A, B, and C, moving in straight lines at constant speeds. The pilot of jet A sees jets B and C moving at the same speed, $c/4$, but while B is moving in parallel, C's trajectory makes an angle of $\pi/4$. What angle does the trajectory of C make with the trajectory of B when viewed from jet B? [5]

Frame of reference of A: (motion along x axis)



we can split C's velocity into two parts

$C_\perp = \frac{c}{4\sqrt{2}}$ and $C_\parallel = \frac{c}{4\sqrt{2}} = \frac{c}{4}\cos\left(\frac{\pi}{4}\right)$

$= \frac{c}{4}\sin\left(\frac{\pi}{4}\right)$

consider ∥ direction (along x)

wrt to A, B is moving along x with speed $v = \frac{c}{4}$ and C is moving with speed $\frac{c}{4\sqrt{2}}$ along x wrt A.

Switch to B's frame. A moving along −ve x with speed $\frac{c}{4}$ $\left(v = -\frac{c}{4}\right)$ and C moving along +x wrt A with $u_x' = \frac{c}{4\sqrt{2}}$

now, $u_x = \dfrac{u_x' + v}{1 + \frac{v u_x'}{c^2}}$

$u_x' \to$ speed of C rel to A (along ∥)
$u_x \to$ speed of C rel to B (∥)
$v \to$ speed of A rel to B. (∥)

$$= \frac{\frac{c}{4\sqrt{2}} + \left(-\frac{c}{4}\right)}{1 + \frac{c^2}{16\sqrt{2}c^2}} = \frac{c\left(\frac{1}{4\sqrt{2}} - \frac{\sqrt{2}}{4\sqrt{2}}\right)}{\left(\frac{16\sqrt{2}+1}{16\sqrt{2}}\right)} = \frac{(1-\sqrt{2})}{4(16\sqrt{2}+1)}c. = \frac{-0.414}{94.05} = 4.4 \times 10^{-3}$$

consider ⊥ direction (along y)

B's frame $u_y = \dfrac{u_y' \sqrt{1 - v^2/c^2}}{1 + v u_x / c^2}$

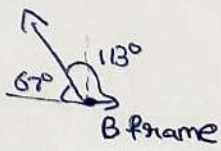
$u_y' \to$ speed of C rel to A (⊥)
$u_y \to$ C rel to B (⊥)
$v \to$ A rel to B (∥)
$u_x \to$ C rel to B (∥)

$$= \frac{\frac{c}{4\sqrt{2}}\sqrt{1 - \frac{1}{16}}}{1 + \frac{c^2}{16\sqrt{2}c^2}}$$

$$= \frac{\left(\frac{\sqrt{15}}{16\sqrt{2}}\right)c}{\left(\frac{1+16\sqrt{2}}{16\sqrt{2}}\right)} = \frac{\sqrt{15}c}{1+16\sqrt{2}} = \frac{3.87c}{23.6} = 1.6 \times 10^{-1}c$$

angle made $\to \theta$ $\tan\theta = \frac{u_y}{u_x} = \frac{\sqrt{15}}{4\,16\sqrt{2}} \Big/ \frac{(1-\sqrt{2})}{4\sqrt{2}}$

$$= \frac{\sqrt{15}}{4(1-\sqrt{2})} = \frac{3.87}{-1.64} = -2.36$$



$\theta = \tan^{-1}(-2.36) = -67^\circ \sim -0.37$ radians.

113 degrees along axis of B
0.62 radians — Ans.